

Abstract No.

Programming glioblastoma subtype identity using synthetic genetic tracing

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Glioblastoma is a highly aggressive brain tumor characterized by its intra- and inter-patient heterogeneity, including non-genetic cell state diversity. The development of accurate glioblastoma models remains a challenge. To screen for factors that enhance the cellular and molecular fidelity of the models, we utilized our synthetic reporters for the classical, proneural, and mesenchymal molecular subtypes in combination with a custom library of supplements, small molecules and biologicals in a high-content screening platform. Using this approach, we have established a defined standardized differentiation protocol imparting or restoring glioblastoma identity applicable to engineered, established and patient-derived cell lines. Our high-fidelity glioblastoma models differentially respond to drugs, highlighting their potential to inform precision treatment strategies.



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